



CYPRUS INTERNATIONAL UNIVERSITY

CMPE343

Database Management Systems and Programming I

Instructor: Dr. Mohammad

Ercan Airport database management system

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Group Members:

- Abubacar Sumalgy Junior - [22216839]
- Maham Samie - [22300391]

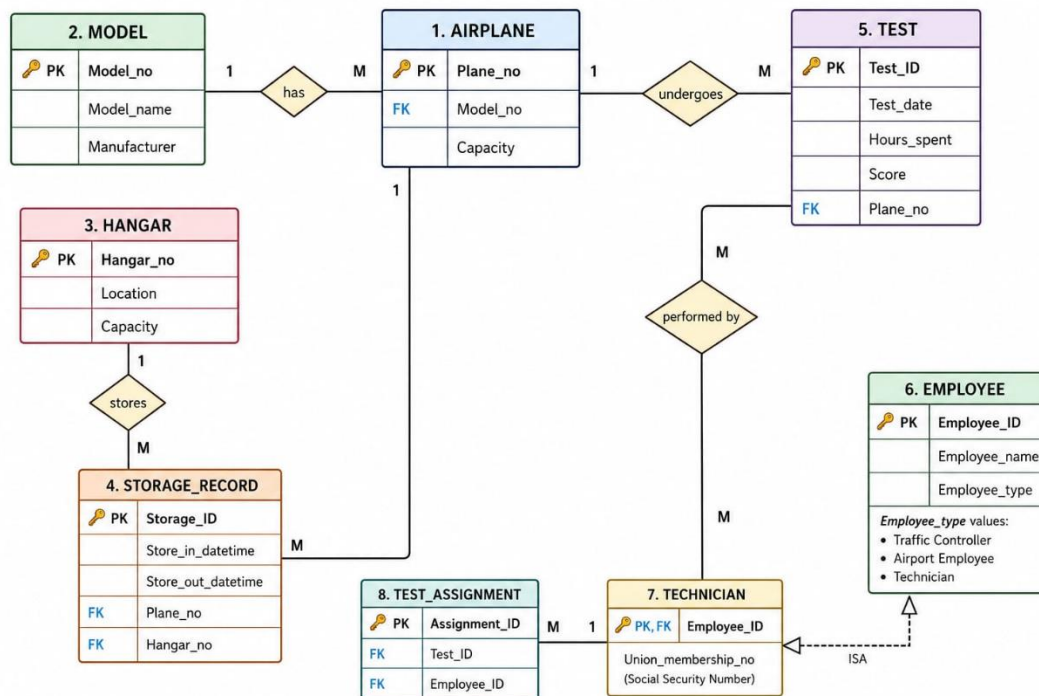
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Introduction

This project represents Ercan Airport Management database system designed to organize and manage information related to airplanes, hangars, employees, technicians, and aircraft testing procedures. The system stores important details such as airplane models and capacities, hangar locations, storage records, maintenance tests, and employee information. It also illustrates the relationships between different entities, including how airplanes undergo tests, how technicians perform those tests, and how airplanes are stored in hangars over time.

Er Diagram

The ER diagram represents the entities in the system and the relationships between them, including customers, menu items, orders, staff, dining tables, payments, and menu categories.



Relational model

Data Definition Language (DDL)

The database scheme was implemented using SQL and includes primary keys, foreign keys, constraints, and default values to ensure data integrity. The scheme defines related tables and enforces relationships between them using foreign key constraints.

The DDL code is:

DROP TABLE IF EXISTS technician_expertise;

DROP TABLE IF EXISTS testing_event;

DROP TABLE IF EXISTS hangar_assignment;

DROP TABLE IF EXISTS technician;

DROP TABLE IF EXISTS traffic_controller;

DROP TABLE IF EXISTS employee_union;

DROP TABLE IF EXISTS airport_employee;

DROP TABLE IF EXISTS union_;

DROP TABLE IF EXISTS airplane;

DROP TABLE IF EXISTS hangar;

DROP TABLE IF EXISTS test;

DROP TABLE IF EXISTS plane_model;

CREATE TABLE union_ (
 union_id INT PRIMARY KEY AUTOINCREMENT,
 union_name VARCHAR(100) NOT NULL UNIQUE,
 contact_info VARCHAR(200),
 established_date DATE,
 monthly_fee NUMERIC(8,2) DEFAULT 0

);

```
CREATE TABLE airport_employee (  
    ssn VARCHAR(11) PRIMARY KEY,  
    full_name VARCHAR(120) NOT NULL,  
    address VARCHAR(250),  
    phone VARCHAR(20),  
    hire_date DATE NOT NULL DEFAULT CURRENT_DATE,  
    birth_date DATE,  
    gender CHAR(1),  
    email VARCHAR(100) UNIQUE  
);
```

```
CREATE TABLE employee_union (  
    ssn VARCHAR(11) NOT NULL,  
    union_id INTEGER NOT NULL,  
    membership_number VARCHAR(50) NOT NULL UNIQUE,  
    join_date DATE NOT NULL DEFAULT CURRENT_DATE,  
    is_active BOOLEAN DEFAULT TRUE,  
    PRIMARY KEY (ssn, union_id),  
    FOREIGN KEY (ssn) REFERENCES airport_employee(ssn) ON UPDATE CASCADE  
    ON DELETE CASCADE,  
    FOREIGN KEY (union_id) REFERENCES union_(union_id) ON UPDATE CASCADE  
    ON DELETE RESTRICT  
);
```

```
CREATE TABLE technician (  
    ssn VARCHAR(11) PRIMARY KEY,
```

```
specialization VARCHAR(100),  
years_of_experience INTEGER DEFAULT 0,  
certification_level VARCHAR(20),  
hourly_rate NUMERIC(10,2) DEFAULT 0,  
FOREIGN KEY (ssn) REFERENCES airport_employee(ssn) ON UPDATE CASCADE  
ON DELETE CASCADE  
);
```

```
CREATE TABLE traffic_controller (  
    ssn VARCHAR(11) PRIMARY KEY,  
    last_medical_exam_date DATE NOT NULL,  
    tower_assignment VARCHAR(50),  
    certification_number VARCHAR(50) UNIQUE,  
    shift_schedule VARCHAR(50),  
    FOREIGN KEY (ssn) REFERENCES airport_employee(ssn) ON UPDATE CASCADE  
ON DELETE CASCADE  
);
```

```
CREATE TABLE plane_model (  
    model_no VARCHAR(30) PRIMARY KEY,  
    manufacturer VARCHAR(60) NOT NULL,  
    type VARCHAR(50) NOT NULL,  
    typical_capacity INT,  
    max_range_km INT,  
    cruise_speed_kmh INT  
);
```

```
CREATE TABLE technician_expertise (  

```

technician_ssn **VARCHAR(11) NOT NULL**,
model_no **VARCHAR(30) NOT NULL**,
expertise_level **VARCHAR(20) DEFAULT 'Certified'**,
certification_date **DATE NOT NULL DEFAULT CURRENT_DATE**,
PRIMARY KEY (technician_ssn, model_no),
FOREIGN KEY (technician_ssn) **REFERENCES** technician(ssn) **ON UPDATE**
CASCADE ON DELETE CASCADE,
FOREIGN KEY (model_no) **REFERENCES** plane_model(model_no) **ON UPDATE**
CASCADE ON DELETE CASCADE
);

CREATE TABLE hangar (
hangar_no **VARCHAR(20) PRIMARY KEY**,
location **VARCHAR(100) NOT NULL**,
capacity **INT**,
has_fire_suppression **BOOLEAN DEFAULT FALSE**,
has_lighting **BOOLEAN DEFAULT TRUE**
);

CREATE TABLE airplane (
plane_no **VARCHAR(20) PRIMARY KEY**,
model_no **VARCHAR(30) NOT NULL**,
current_hangar_no **VARCHAR(20)**,
status **VARCHAR(20) DEFAULT 'Active'**,
acquisition_date **DATE** ,
last_maintenance_date **DATE**,
total_flight_hours **INT**,

**FOREIGN KEY (model_no) REFERENCES plane_model(model_no) ON UPDATE
CASCADE ON DELETE RESTRICT,**

**FOREIGN KEY (current_hangar_no) REFERENCES hangar(hangar_no) ON UPDATE
CASCADE ON DELETE SET NULL**

);

CREATE TABLE hangar_assignment (

assignment_id INT PRIMARY KEY AUTOINCREMENT,

plane_no VARCHAR(20) NOT NULL,

hangar_no VARCHAR(20) NOT NULL,

in_date DATE NOT NULL,

in_time TIME NOT NULL,

out_date DATE,

out_time TIME,

reason VARCHAR(100),

**FOREIGN KEY (plane_no) REFERENCES airplane(plane_no) ON UPDATE CASCADE
ON DELETE CASCADE,**

**FOREIGN KEY (hangar_no) REFERENCES hangar(hangar_no) ON UPDATE
CASCADE ON DELETE RESTRICT**

);

CREATE TABLE test (

test_id INT PRIMARY KEY AUTOINCREMENT,

test_name VARCHAR(100) NOT NULL UNIQUE,

description TEXT,

duration_hours NUMERIC(5,2) NOT NULL,

frequency_days INT,

passing_score INT

);

```
CREATE TABLE testing_event (  
    event_id INT PRIMARY KEY AUTOINCREMENT,  
    plane_no VARCHAR(20) NOT NULL,  
    technician_ssn VARCHAR(11) NOT NULL,  
    test_id INT NOT NULL,  
    test_date DATE NOT NULL,  
    test_time TIME NOT NULL,  
    hours_spent NUMERIC (5,2) NOT NULL,  
    score INT,  
    remarks TEXT,  
  
    FOREIGN KEY (plane_no) REFERENCES airplane(plane_no) ON UPDATE CASCADE  
ON DELETE RESTRICT,  
  
    FOREIGN KEY (technician_ssn) REFERENCES technician(ssn) ON UPDATE  
CASCADE ON DELETE RESTRICT,  
  
    FOREIGN KEY (test_id) REFERENCES test(test_id) ON UPDATE CASCADE ON  
DELETE RESTRICT,  
  
    UNIQUE (plane_no, test_id, test_date)  
);
```

Data Manipulation Language (DML)

Sample data was inserted into all tables using SQL INSERT statements. UPDATE and DELETE operations were also performed to demonstrate data modification and removal.

The DML code is:

INSERT INTO union_ (union_name, contact_info, established_date, monthly_fee) VALUES

('Airport Workers Union TRNC', 'contact@awu.trnc', '1985-03-15', 50.00),
('Aviation Technicians Guild', 'info@atg.trnc', '1990-07-22', 75.00),
('Air Traffic Controllers Association', 'atca@ercan.trnc', '1978-11-10', 60.00);

INSERT INTO plane_model (model_no, manufacturer, type, typical_capacity, max_range_km, cruise_speed_kmh) VALUES

('B737-800', 'Boeing', 'Narrow-body', 189, 5400, 842),
('A320-200', 'Airbus', 'Narrow-body', 180, 6150, 828),
('B777-300ER', 'Boeing', 'Wide-body', 354, 13650, 905),
('B787-9', 'Boeing', 'Wide-body', 290, 14140, 903);

INSERT INTO airport_employee (ssn, full_name, address, phone, hire_date, birth_date, gender, email) VALUES

('12345678901', 'Hakan Şükür', 'Lefkosa, TRNC', '+90-533-111-2233', '2010-06-15', '1971-09-01', 'M', 'hakan.sukur@ercan.trnc'),
('23456789012', 'Arda Turan', 'Girne, TRNC', '+90-542-222-3344', '2012-09-20', '1987-01-30', 'M', 'arda.turan@ercan.trnc'),
('34567890123', 'Burak Yılmaz', 'Guzelyurt, TRNC', '+90-533-333-4455', '2015-03-10', '1985-07-15', 'M', 'burak.yilmaz@ercan.trnc'),
('56789012345', 'Tuncay Şanlı', 'Girne, TRNC', '+90-533-555-6677', '2008-11-01', '1982-01-16', 'M', 'tuncay.sanli@ercan.trnc'),
('67890123456', 'Fatih Terim', 'Lefkosa, TRNC', '+90-542-666-7788', '2020-06-20', '1953-09-04', 'M', 'fatih.terim@ercan.trnc'),

('78901234567', 'Rüştü Reçber', 'Magusa, TRNC', '+90-533-777-8899', '2005-09-25', '1973-05-10', 'M', 'rustu.recber@ercan.trnc'),

('89012345678', 'Hamit Altıntop', 'Lefkosa, TRNC', '+90-548-888-9900', '2019-03-18', '1982-12-08', 'M', 'hamit.altintop@ercan.trnc'),

('01234567890', 'Gökhan Gönül', 'Lefkosa, TRNC', '+90-542-111-2233', '2016-12-05', '1985-01-04', 'M', 'gokhan.gonul@ercan.trnc');

INSERT INTO technician (ssn, specialization, years_of_experience, certification_level, hourly_rate) **VALUES**

('12345678901', 'Engine Specialist', 14, 'Master', 85.00),

('23456789012', 'Avionics Engineer', 12, 'Senior', 70.00),

('34567890123', 'Airframe Specialist', 9, 'Senior', 65.00),

('56789012345', 'Maintenance Engineer', 16, 'Master', 90.00);

INSERT INTO traffic_controller (ssn, last_medical_exam_date, tower_assignment, certification_number, shift_schedule) **VALUES**

('67890123456', '2025-06-15', 'Main Tower', 'ATC-TRNC-001', 'Morning Shift'),

('78901234567', '2025-05-20', 'Main Tower', 'ATC-TRNC-002', 'Night Shift'),

('89012345678', '2025-04-10', 'Approach Control', 'ATC-TRNC-003', 'Evening Shift');

INSERT INTO employee_union (ssn, union_id, membership_number, join_date, is_active) **VALUES**

('12345678901', 2, 'ATG-1001', '2010-07-01', 1),

('23456789012', 2, 'ATG-1002', '2012-10-05', 1),

('34567890123', 2, 'ATG-1003', '2015-04-20', 1),

('56789012345', 2, 'ATG-1005', '2008-12-15', 1),

('67890123456', 3, 'ATCA-2001', '2020-07-15', 1),
('78901234567', 3, 'ATCA-2002', '2005-10-01', 1),
('89012345678', 3, 'ATCA-2003', '2019-04-01', 1);

INSERT INTO technician_expertise (technician_ssn, model_no, expertise_level, certification_date) **VALUES**

('12345678901', 'B737-800', 'Master', '2012-06-20'),
('12345678901', 'B787-9', 'Expert', '2016-08-15'),
('23456789012', 'A320-200', 'Expert', '2014-09-10'),
('34567890123', 'B737-800', 'Expert', '2016-03-15'),
('34567890123', 'B777-300ER', 'Certified', '2018-07-22'),
('56789012345', 'B777-300ER', 'Master', '2010-12-01'),
('56789012345', 'B787-9', 'Expert', '2015-04-18');

INSERT INTO hangar (hangar_no, location, capacity, has_fire_suppression, has_lighting) **VALUES**

('H-01', 'North Section - Terminal A - Ercan Airport', 8, 1, 1),
('H-02', 'South Section - Maintenance Area - Ercan Airport', 12, 1, 1),
('H-03', 'East Section - Cargo Area - Ercan Airport', 6, 0, 1);

INSERT INTO airplane (plane_no, model_no, current_hangar_no, status, acquisition_date, last_maintenance_date, total_flight_hours) **VALUES**

('TC-ABC', 'B737-800', 'H-01', 'Active', '2018-03-10', '2025-11-15', 8500),
('TC-DEF', 'A320-200', 'H-02', 'Active', '2019-06-20', '2025-11-20', 6200),

('TC-GHI', 'B777-300ER', 'H-03', 'In Maintenance', '2016-09-15', '2025-10-25', 18500),

('TC-JKL', 'B787-9', 'H-01', 'Active', '2021-04-05', '2025-11-05', 3100);

INSERT INTO hangar_assignment (plane_no, hangar_no, in_date, in_time, out_date, out_time, reason) **VALUES**

('TC-ABC', 'H-01', '2024-01-10', '08:00:00', '2024-06-15', '14:30:00', 'Regular Parking'),

('TC-ABC', 'H-01', '2024-06-15', '15:00:00', NULL, NULL, 'Return to Service'),

('TC-DEF', 'H-02', '2024-03-01', '09:00:00', NULL, NULL, 'Regular Parking'),

('TC-GHI', 'H-03', '2024-02-15', '07:00:00', NULL, NULL, 'Major Overhaul'),

('TC-JKL', 'H-01', '2024-05-01', '11:00:00', NULL, NULL, 'Regular Parking');

INSERT INTO test (test_name, description, duration_hours, frequency_days, passing_score) **VALUES**

('Engine Performance Test', 'Comprehensive engine thrust and efficiency test', 4.5, 180, 85),

('Avionics Systems Check', 'Testing navigation and communication systems', 3.0, 90, 90),

('Emergency Systems Test', 'Testing all emergency equipment', 2.0, 180, 95),

('Landing Gear Operation', 'Full cycle landing gear deployment test', 2.5, 120, 85);

INSERT INTO testing_event (plane_no, technician_ssn, test_id, test_date, test_time, hours_spent, score, remarks) **VALUES**

('TC-ABC', '12345678901', 1, '2025-11-15', '09:00:00', 4.0, 92, 'Excellent performance'),

('TC-ABC', '23456789012', 2, '2025-11-10', '13:30:00', 2.5, 88, 'Minor calibration needed'),
('TC-DEF', '34567890123', 4, '2025-11-08', '14:00:00', 2.0, 90, 'Smooth operation'),
('TC-GHI', '56789012345', 1, '2025-10-25', '08:30:00', 4.5, 65, 'Engine needs overhaul'),
('TC-GHI', '56789012345', 3, '2025-10-26', '11:00:00', 2.0, 82, 'Emergency systems OK'),
('TC-JKL', '12345678901', 2, '2025-11-01', '15:00:00', 2.8, 95, 'All systems nominal');

UPDATE airplane **SET** status = 'Active', last_maintenance_date = **DATE**('now')
WHERE plane_no = 'TC-GHI';

UPDATE airplane **SET** current_hangar_no = 'H-02' **WHERE** plane_no = 'TC-DEF';

INSERT INTO hangar_assignment (plane_no, hangar_no, in_date, in_time, reason) **VALUES** ('TC-DEF', 'H-02', **DATE**('now'), **TIME**('now'), 'Relocation for maintenance');

UPDATE technician **SET** hourly_rate = hourly_rate * **1.10** **WHERE** certification_level = 'Master';

UPDATE traffic_controller **SET** last_medical_exam_date = **DATE**('now')
WHERE ssn = '78901234567';

UPDATE test **SET** passing_score = passing_score + 5 **WHERE** test_name = 'Engine Performance Test';

UPDATE airplane **SET** total_flight_hours = total_flight_hours + 500 **WHERE** status = 'Active';

DELETE FROM hangar_assignment **WHERE** out_date IS NOT NULL AND out_date < **DATE**('now', '-2 years');

SQL queries:

Q1: List all airplanes with their model details and current hangar

Uses: Join

```
SELECT
    a.plane_no,
    pm.manufacturer,
    pm.type,
    a.status,
    a.total_flight_hours,
    h.location AS hangar_location
FROM airplane a
JOIN plane_model pm ON a.model_no = pm.model_no
JOIN hangar h ON a.current_hangar_no = h.hangar_no;
```

plane_no	manufacturer	type	status	total_flight_hours	hangar_location
TC-ABC	Boeing	Narrow-body	Active	8500	North Section - Terminal A-...
TC-DEF	Airbus	Narrow-body	Active	6200	South Section - Maintenanc...
TC-GHI	Boeing	Wide-body	In Maintenance	18500	East Section - Cargo Area -...
TC-JKL	Boeing	Wide-body	Active	3100	North Section - Terminal A-...

Q2: Show all testing events with airplane, technician name and test name

Uses: Join(multiple)

SELECT

```
te.event_id,  
te.plane_no,  
ae.full_name AS technician_name,  
t.test_name,  
te.test_date,  
te.hours_spent,  
te.score,  
te.remarks
```

FROM testing_event te

JOIN technician tech ON te.technician_ssn = tech.ssn

JOIN airport_employee ae ON tech.ssn = ae.ssn

JOIN test t ON te.test_id = t.test_id

ORDER BY te.test_date DESC;

event_id	plane_no	technician_n...	test_name	test_date	hours_spent	score	remarks
1	TC-ABC	Hakan Şükür	Engine Perfor...	2025-11-15	4	92	Excellent performance
2	TC-ABC	Arda Turan	Avionics Syste...	2025-11-10	2.5	88	Minor calibration error
3	TC-DEF	Burak Yılmaz	Landing Gear ...	2025-11-08	2	90	Smooth operation
5	TC-GHI	Tuncay Şanlı	Emergency Sy...	2025-10-26	2	82	Emergency system f...
4	TC-GHI	Tuncay Şanlı	Engine Perfor...	2025-10-25	4.5	65	Engine needs mainte...

Q3: Total hours each technician has spent on tests

Uses: Join, Group By

```
SELECT
    ae.full_name AS technician_name,
    tech.specialization,
    COUNT(te.event_id) AS total_tests,
    SUM(te.hours_spent) AS total_hours_spent,
    ROUND(AVG(te.hours_spent), 2) AS avg_hours_per_test
FROM testing_event te
JOIN technician tech ON te.technician_ssn = tech.ssn
JOIN airport_employee ae ON tech.ssn = ae.ssn
GROUP BY ae.full_name, tech.specialization
ORDER BY total_hours_spent DESC;
```

technician_name	specialization	total_tests	total_hours_spent	avg_hours_per_test
Tuncay Şanlı	Maintenance Engineer	2	6.5	3.25
Hakan Şükür	Engine Specialist	1	4	4
Arda Turan	Avionics Engineer	1	2.5	2.5
Burak Yılmaz	Airframe Specialist	1	2	2

Q4: Average test score per airplane

Uses: Join, Group By

```
SELECT
    a.plane_no,
    pm.manufacturer,
    pm.type,
    COUNT(te.event_id) AS number_of_tests,
    ROUND(AVG(te.score), 2) AS average_score,
    MIN(te.score) AS lowest_score,
    MAX(te.score) AS highest_score
FROM airplane a
JOIN plane_model pm ON a.model_no = pm.model_no
LEFT JOIN testing_event te ON a.plane_no = te.plane_no
GROUP BY a.plane_no, pm.manufacturer, pm.type
ORDER BY average_score ASC;
```

plane_no	manufacturer	type	number_of_tests	average_score	lowest_score	highest_score
TC-JKL	Boeing	Wide-body	0	NULL	NULL	NULL
TC-GHI	Boeing	Wide-body	2	73.5	65	82
TC-ABC	Boeing	Narrow-body	2	90	88	92
TC-DEF	Airbus	Narrow-body	1	90	90	90

Q5: Airplanes that have failed at least one test

Uses: Join, Subquery

```
SELECT
    a.plane_no,
    pm.manufacturer,
    t.test_name,
    te.score,
    t.passing_score,
    te.test_date,
    ae.full_name AS tested_by
FROM testing_event te
JOIN airplane a ON te.plane_no = a.plane_no
JOIN plane_model pm ON a.model_no = pm.model_no
JOIN test t ON te.test_id = t.test_id
JOIN technician tech ON te.technician_ssn = tech.ssn
JOIN airport_employee ae ON tech.ssn = ae.ssn
WHERE te.score < (
    SELECT t2.passing_score
    FROM test t2
    WHERE t2.test_id = te.test_id
)
ORDER BY te.score ASC;
```

plane_no	manufacturer	test_name	score	passing_score	test_date	tested_by
TC-ABC	Boeing	Avionics Systems...	88	90	2025-11-10	Arda Turan
TC-GHI	Boeing	Engine Performa...	65	85	2025-10-25	Tuncay Şanlı
TC-GHI	Boeing	Emergency Syste...	82	95	2025-10-26	Tuncay Şanlı

Q6: Number of airplanes currently stored in each hangar

Uses: Join, Group By

```
SELECT
    h.hangar_no,
    h.location,
    h.capacity AS max_capacity,
    COUNT(a.plane_no) AS planes_currently_stored,
    (h.capacity - COUNT(a.plane_no)) AS available_spots
FROM hangar h
LEFT JOIN airplane a ON h.hangar_no = a.current_hangar_no
GROUP BY h.hangar_no, h.location, h.capacity
ORDER BY planes_currently_stored DESC;
```

hangar_no	location	max_capacity	planes_currently_stored	available_spots
H-01	North Section - Terminal A ...	8	2	6
H-02	South Section - Maintenanc...	12	1	11
H-03	East Section - Cargo Area ...	6	1	5

Q7: Traffic controllers whose medical exam is older than 1 year

Uses: Join

```
SELECT
    ae.full_name,
    tc.tower_assignment,
    tc.shift_schedule,
    tc.last_medical_exam_date,
    ROUND(JULIANDAY('now') - JULIANDAY(tc.last_medical_exam_date)) AS days_since_exam
FROM traffic_controller tc
JOIN airport_employee ae ON tc.ssn = ae.ssn
WHERE tc.last_medical_exam_date < DATE('now', '-1 year')
ORDER BY tc.last_medical_exam_date ASC;
```

full_name	tower_assignment	shift_schedule	last_medical_exam_date	days_since_exam
Hamit Altıntop	Approach Control	Evening Shift	2025-04-10	410
Rüştü Reçber	Main Tower	Night Shift	2025-05-20	370

Q8:List all technicians and the plane models they are certified for

Uses: Join, Group By

```

SELECT
    ae.full_name AS technician_name,
    tech.specialization,
    tech.certification_level,
    GROUP_CONCAT(te.model_no SEPARATOR ', ') AS certified_models,
    COUNT(te.model_no) AS number_of_certifications
FROM technician tech
JOIN airport_employee ae ON tech.ssn = ae.ssn
LEFT JOIN technician_expertise te ON tech.ssn = te.technician_ssn
GROUP BY ae.full_name, tech.specialization, tech.certification_level
ORDER BY number_of_certifications DESC;

```

technician_name	specialization	certification_level	certified_models	number_of_certifications
Burak Yılmaz	Airframe Specialist	Senior	B737-800, B777-300ER	2
Hakan Şükür	Engine Specialist	Master	B737-800, B787-9	2
Tuncay Şanlı	Maintenance Engineer	Master	B777-300ER, B787-9	2
Arda Turan	Avionics Engineer	Senior	A320-200	1

Q9: Technicians certified for more than one plane model

Use: Join, Group By, Subquery

```

SELECT
    ae.full_name AS technician_name,
    tech.specialization,
    COUNT(te.model_no) AS number_of_models
FROM technician tech
JOIN airport_employee ae ON tech.ssn = ae.ssn
JOIN technician_expertise te ON tech.ssn = te.technician_ssn
GROUP BY ae.full_name, tech.specialization
HAVING COUNT(te.model_no) > 1
ORDER BY number_of_models DESC;

```

technician_name	specialization	number_of_models
Burak Yılmaz	Airframe Specialist	2
Hakan Şükür	Engine Specialist	2
Tuncay Şanlı	Maintenance Engineer	2

Q10: Full hangar history for each airplane with formatted dates

Uses: Join, DATE_FORMAT

```
SELECT
    ha.plane_no,
    h.location AS hangar_location,
    STRFTIME('%d/%m/%Y', ha.in_date) AS checked_in,
    CASE
        WHEN ha.out_date IS NULL THEN 'Still in hangar'
        ELSE STRFTIME('%d/%m/%Y', ha.out_date)
    END AS checked_out,
    ha.reason
FROM hangar_assignment ha
JOIN hangar h ON ha.hangar_no = h.hangar_no
ORDER BY ha.plane_no, ha.in_date ASC;
```

plane_no	hangar_location	checked_in	checked_out	reason
TC-ABC	North Section - Terminal A ...	10/01/2024	15/06/2024	Regular Parking
TC-ABC	North Section - Terminal A ...	15/06/2024	Still in hangar	Return to Service
TC-DEF	South Section - Maintenanc...	01/03/2024	Still in hangar	Regular Parking
TC-GHI	East Section - Cargo Area ...	15/02/2024	Still in hangar	Major Overhaul
TC-JKL	North Section - Terminal A ...	01/05/2024	Still in hangar	Regular Parking

Q11: Most frequently used tests

Uses: Join, Group By

```

SELECT
    t.test_name,
    t.passing_score,
    t.duration_hours,
    COUNT(te.event_id) AS times_performed,
    ROUND(AVG(te.score), 2) AS average_score_achieved
FROM test t
LEFT JOIN testing_event te ON t.test_id = te.test_id
GROUP BY t.test_name, t.passing_score, t.duration_hours
ORDER BY times_performed DESC;

```

test_name	passing_score	duration_hours	times_performed	average_score_achieved
Engine Performance Test	85	4.5	2	78.5
Avionics Systems Check	90	3	1	88
Emergency Systems Test	95	2	1	82
Landing Gear Operation	85	2.5	1	90

Q12: Employees who are NOT in any union

Uses: Subquery

```

SELECT
    ae.ssn,
    ae.full_name,
    ae.hire_date,
    ae.email
FROM airport_employee ae
WHERE ae.ssn NOT IN (
    SELECT eu.ssn
    FROM employee_union eu
)
ORDER BY ae.hire_date;

```

ssn	full_name	hire_date	email
01234567890	Gökhan Gönül	2016-12-05	gokhan.gonul@ercan.trnc

Q13: Monthly testing activity report

Uses: Group by, Date_Format

```
SELECT
    ae.ssn,
    ae.full_name,
    ae.hire_date,
    ae.email
FROM airport_employee ae
WHERE ae.ssn NOT IN (
    SELECT eu.ssn
    FROM employee_union eu
)
ORDER BY ae.hire_date;
SELECT
    STRFTIME('%Y', te.test_date) AS year,
    STRFTIME('%m', te.test_date) AS month,
    COUNT(te.event_id) AS total_tests_conducted,
    ROUND(AVG(te.score), 2) AS average_score,
    SUM(te.hours_spent) AS total_hours_spent
FROM testing_event te
GROUP BY STRFTIME('%Y', te.test_date), STRFTIME('%m', te.test_date)
ORDER BY year DESC, month DESC;
```

year	month	total_tests_conducted	average_score	total_hours_spent
2025	October	2	73.5	6.5
2025	November	3	90	8.5

Q14: Airplanes that have never been tested

Uses: Subquery

```

SELECT
    a.plane_no,
    pm.manufacturer,
    pm.type,
    a.status,
    a.acquisition_date,
    a.total_flight_hours
FROM airplane a
JOIN plane_model pm ON a.model_no = pm.model_no
WHERE a.plane_no NOT IN (
    SELECT DISTINCT te.plane_no
    FROM testing_event te
)
ORDER BY a.acquisition_date;

```

plane_no	manufacturer	type	status	acquisition_date	total_flight_hours
TC-JKL	Boeing	Wide-body	Active	2021-04-05	3100

Q15: Technician cost report; total pay per technician based on hours spent

Uses: Join, Group By

```

SELECT
    ae.full_name AS technician_name,
    tech.certification_level,
    tech.hourly_rate,
    SUM(te.hours_spent) AS total_hours_worked,
    ROUND(SUM(te.hours_spent) * tech.hourly_rate, 2) AS total_pay_due
FROM testing_event te
JOIN technician tech ON te.technician_ssn = tech.ssn
JOIN airport_employee ae ON tech.ssn = ae.ssn
GROUP BY ae.full_name, tech.certification_level, tech.hourly_rate
ORDER BY total_pay_due DESC;

```

technician_name	certification_level	hourly_rate	total_hours_worked	total_pay_due
Tuncay Şanlı	Master	90	6.5	585
Hakan Şükür	Master	85	4	340
Arda Turan	Senior	70	2.5	175
Burak Yılmaz	Senior	65	2	130

SQL queries table: